

Why small tanh slopes make readout gradient descent slow

Two-sided finite-network eigenvalue bounds and training time

Section 3 source note 2 — 25 September 2026

An accurate readout can exist even when ordinary gradient descent needs an impractical number of updates to find it. We study a tanh network with frozen centers and a common slope γ . The argument has four steps: derive gradient descent's dependence on normalized kernel eigenvalues; replace the center sum by an explicit integral; bound the finite eigenvalues above and below using that integral and controlled corrections; and substitute the ratio interval into the target's error formula. Small gamma attenuates high frequencies in the integral. The finite-network bounds quantify the resulting small learning rates in the geometry we test.

The theorem is an interval evaluated at each gamma, with no starting gamma or reference eigenvalue. Its upper ratio endpoints give necessary training times; its lower endpoints give sufficient times. This does not assert that every actual ratio increases for every geometry. The mathematical inequalities are exact; the figures are checked floating-point evaluations, not interval-arithmetic certificates. All errors and times concern the specified training samples and raw readout coordinates.

1 Frozen geometry: the normalized eigenvalues determine convergence

For $m \geq 2$ training inputs x_a and W tanh centers c_j , including halo centers, define $B_\gamma \in \mathbb{R}^{m \times (W+1)}$ by

$$(B_\gamma)_{aj} = m^{-1/2} \tanh(\gamma(x_a - c_j)), \quad (B_\gamma)_{a0} = m^{-1/2}.$$

Column zero is the bias feature. With $y_a = f(x_a)/\sqrt{m}$, the readout coefficients v minimize $L(v) = \frac{1}{2} \|B_\gamma v - y\|^2$. Geometry is fixed throughout training. Writing $r_n = B_\gamma v_n - y$, an ordinary gradient descent update gives

$$\begin{aligned} v_{n+1} &= v_n - \eta_\gamma B_\gamma^\top r_n, \\ r_{n+1} &= B_\gamma(v_n - \eta_\gamma B_\gamma^\top r_n) - y = (I - \eta_\gamma K_\gamma) r_n, \quad K_\gamma = B_\gamma B_\gamma^\top. \end{aligned} \tag{1}$$

Starting at $v_0 = 0$ gives $r_0 = -y$. Order the kernel eigenvalues as $\lambda_1 \geq \dots \geq \lambda_m \geq 0$, with orthonormal eigenvectors u_i . Throughout, use

$$\eta_\gamma = \frac{1}{2\lambda_1(\gamma)}, \quad \rho_i(\gamma) = \frac{\lambda_i(\gamma)}{\lambda_1(\gamma)}.$$

Each eigenvector component then satisfies

$$u_i^\top r_n = -u_i^\top y (1 - \rho_i/2)^n. \tag{2}$$

Multiplying the entire kernel by a scalar leaves these factors unchanged. What matters is the ratio ρ_i , together with the target component along u_i . Define its squared-norm fraction by $p_i(\gamma) = |u_i(\gamma)^\top y|^2 / \|y\|^2$. The exact relative training error is

$$E_\gamma(n)^2 := \frac{\|r_n\|^2}{\|y\|^2} = \sum_{i=1}^m p_i(\gamma) (1 - \rho_i(\gamma)/2)^{2n}. \quad (3)$$

The weights sum to one. A zero eigenvalue contributes an irreducible component; a small positive eigenvalue contributes a representable component that decays slowly. The next two sections bound the ratios. Section 4 uses those bounds in (3), retaining the actual target weights.

2 A center integral separates the explicit matrix from the finite corrections

The actual kernel entries are

$$(K_\gamma)_{ab} = \frac{1}{m} \left[1 + \sum_{j=1}^W g_{ab}(c_j) \right], \quad g_{ab}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c)).$$

We now consider the continuous version K_γ^{int} , taking an integral inner product over functions of the center coordinate c . The sample locations remain fixed. Assume consecutive lattice centers with spacing h ; their cells cover $I_c = [c_1 - h/2, c_W + h/2]$, of length Wh . Then

$$(K_\gamma^{\text{int}})_{ab} = \frac{1}{m} \left[1 + \frac{1}{h} \int_{I_c} g_{ab}(c) dc \right], \quad E_\gamma^{\text{grid}} = K_\gamma - K_\gamma^{\text{int}}. \quad (4)$$

The factor $1/h$ matches the center density. Writing $g_{ab} = 1 - (1 - g_{ab})$, the constant integral gives Wh , while the whole-line deficit has the explicit value

$$\int_{\mathbb{R}} (1 - g_{ab}(c)) dc = 2d_{ab} \coth(\gamma d_{ab}), \quad d_{ab} = x_a - x_b.$$

At $d_{ab} = 0$, the value is $2/\gamma$. Restoring the deficit outside I_c gives the exact decomposition

$$K_\gamma = \underbrace{\frac{W+1}{m} \mathbf{1}\mathbf{1}^\top - \frac{2}{hm} [d_{ab} \coth(\gamma d_{ab})]_{a,b}}_{K_\gamma^{(0)}} + E_\gamma^{\text{halo}} + E_\gamma^{\text{grid}}, \quad (5)$$

where $(E_\gamma^{\text{halo}})_{ab} = (hm)^{-1} \int_{\mathbb{R} \setminus I_c} (1 - g_{ab}(c)) dc$. Appendix A derives the integral.

The explicit matrix has a constant rank-one term and a term depending only on the separation $x_a - x_b$. On equally spaced samples, its entries are constant along each diagonal: this is the Toeplitz structure. It is not a diagonal matrix. The corrections retain the dependence on the finite center interval and the discrete center lattice. We do not assume Fourier waves are eigenvectors of the sampled matrix.

3 The Fourier integral gives a two-sided finite-network ratio bound

Tanh is a step averaged with the density $\rho_\gamma^{\text{av}}(x) = \gamma \operatorname{sech}^2(\gamma x)/2$:

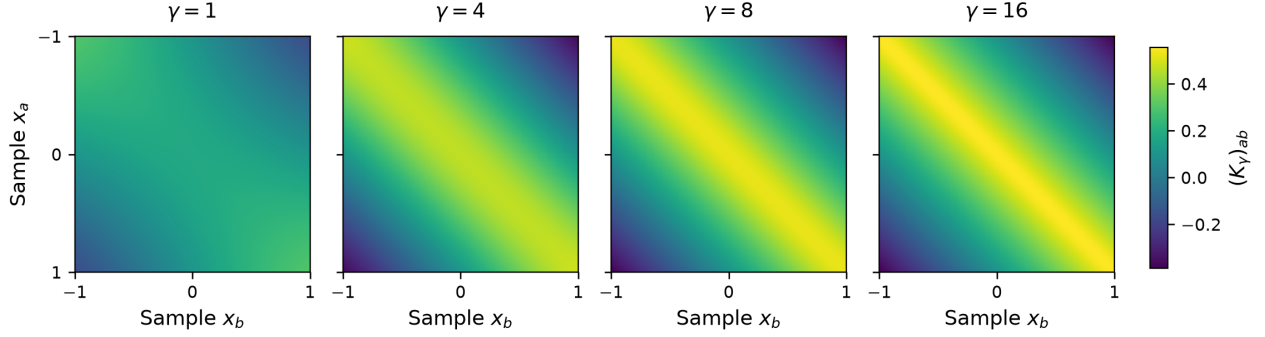


Figure 1: Actual matrices $K_\gamma = B_\gamma B_\gamma^\top$ at $\gamma = 1, 4, 8, 16$, using identical samples and centers and a common color scale. The separation-dependent term produces the diagonal bands. These are full finite matrices, including the bias, grid, and halo contributions.

$$(\rho_\gamma^{\text{av}} * \text{sgn})(x) = 2 \int_{-\infty}^x \rho_\gamma^{\text{av}}(t) dt - 1 = \tanh(\gamma x).$$

With $\hat{f}(\omega) = \int f(x) e^{-i\omega x} dx$, convolution multiplies the transform by

$$M_\gamma(\omega) = \widehat{\rho_\gamma^{\text{av}}}(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1. \quad (6)$$

This averaging acts in both inputs of the feature-product kernel: its two-input Fourier transform is multiplied by $M_\gamma(\omega)M_\gamma(\omega')$ relative to the step kernel. Thus small gamma suppresses high-frequency variation in the kernel entries. The next identity connects that suppression to eigenvalues, without identifying Fourier waves with eigenvectors.

Let $Q \in \mathbb{R}^{m \times (m-1)}$ have orthonormal columns spanning vectors whose entries sum to zero: $Q^\top Q = I$, $Q^\top \mathbf{1} = 0$. This removes one constant direction for the calculation; it places no assumption on the actual eigenvectors. For $t_\gamma(c) = (\tanh(\gamma(x_a - c)))_{a=1}^m$, define

$$H_\gamma := \frac{1}{hm} \int_{\mathbb{R}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q dc = Q^\top K_\gamma^{(0)} Q.$$

The constant tails cancel after projection, so this integral is finite. For any vector $v \in \mathbb{R}^{m-1}$, set $u = Qv$. The following three steps explain its quadratic form:

$$g_\gamma(c) := \sum_a u_a \tanh(\gamma(c - x_a)), \quad v^\top H_\gamma v = \frac{1}{hm} \int |g_\gamma(c)|^2 dc;$$

$$\hat{g}_\gamma(\omega) = M_\gamma(\omega) \frac{2}{i\omega} \sum_a u_a e^{-i\omega x_a};$$

$$\boxed{v^\top H_\gamma v = \frac{2}{\pi hm} \int_{\mathbb{R}} \frac{M_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega x_a} \right|^2 d\omega.} \quad (7)$$

The middle line follows by differentiating the sum of steps; the last is Parseval's identity. Here $u_a = (Qv)_a$ is the entry of a sample-space vector, not a readout coefficient. The apparent singularity at zero is removable because $\sum_a u_a = 0$. Appendix B gives the details.

For a fixed v , all gamma dependence in (7) is in M_γ , and every contribution is nonnegative. Increasing gamma increases M_γ . Therefore $H_{\gamma_b} \succeq H_{\gamma_a}$ for $\gamma_b \geq \gamma_a$, and every ordered eigenvalue of

H_γ is nondecreasing. This statement does not require fixed eigenvectors. Contributions at $|\omega| \gg \gamma$ contain the exponentially small factor $M_\gamma(\omega)^2 \sim 4z^2 e^{-2z}$; those at $|\omega| \ll \gamma$ have a multiplier near one.

To bound the finite eigenvalues, correct the integral for the center lattice and the finite halo. Let G_γ be a finite sum of explicitly computed lattice Fourier corrections, and let T_γ be the outer-product sum of a finite set of omitted exterior lattice centers. Form

$$S_\gamma = H_\gamma + G_\gamma - T_\gamma. \quad (8)$$

The unretained lattice terms have norm at most δ_γ . The remaining omitted-center matrix is positive semidefinite and at most $\tau_\gamma I$. Thus

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^\top K_\gamma Q \preceq S_\gamma + \delta_\gamma I. \quad (9)$$

These allowances are derived analytically from the Fourier and tanh tails; they are not measured differences from the actual kernel. Appendix C specifies the corrections and the bounds. Write $\beta_1(\gamma) \geq \dots \geq \beta_{m-1}(\gamma)$ for the eigenvalues of S_γ . Computing this spectrum is part of the method; (7) is not a closed scalar formula for eigenvalue rank versus gamma.

Theorem 1 (finite-network eigenvalue and ratio interval). Suppose the centers lie on a lattice of spacing h , the common slope is positive, and $S_\gamma, \delta_\gamma, \tau_\gamma$ satisfy (9). Let $0 < \ell_\gamma \leq \lambda_1(K_\gamma) \leq L_\gamma$. For $2 \leq i \leq m-1$, define

$$\underline{\lambda}_i = [\beta_i - \delta_\gamma - \tau_\gamma]_+, \quad \bar{\lambda}_i = \beta_{i-1} + \delta_\gamma, \quad [t]_+ = \max(t, 0).$$

Then

$$\boxed{\begin{aligned} \underline{\lambda}_i &\leq \lambda_i(K_\gamma) \leq \bar{\lambda}_i, \\ \underline{\rho}_i &:= \frac{\underline{\lambda}_i}{L_\gamma} \leq \rho_i \leq \min\left\{1, \frac{\bar{\lambda}_i}{\ell_\gamma}\right\} =: \bar{\rho}_i. \end{aligned}} \quad (10)$$

For $i = m$, use $\underline{\lambda}_m = 0$ and $\bar{\lambda}_m = \beta_{m-1} + \delta_\gamma$. Set $\underline{\rho}_1 = \bar{\rho}_1 = 1$. Known zero eigenvalues can be assigned both endpoints zero. These bounds depend only on the current gamma and geometry, not on a reference slope or old eigenvectors.

The proof is short once (9) is established. Its eigenvalues differ from β_i by the indicated allowances. Removing one dimension gives

$$\lambda_i(K_\gamma) \geq \lambda_i(Q^\top K_\gamma Q), \quad \lambda_i(K_\gamma) \leq \lambda_{i-1}(Q^\top K_\gamma Q).$$

Combining these inequalities gives the different indices in (10). Dividing the lower numerator by an upper denominator, and the upper numerator by a lower denominator, proves the ratio interval. In particular, no finite eigenvector is assumed to have zero mean. We may always take $\ell_\gamma = 1, L_\gamma = W + 1$; the plots use sharper feature-average and block bounds from Appendix D, without using the actual small eigenvalues.

For rank 26 in the default geometry, the evaluated intervals are:

γ	Lower ratio endpoint	Actual ratio	Upper ratio endpoint
4	6.562×10^{-13}	2.088×10^{-12}	2.262×10^{-12}
8	7.719×10^{-8}	1.487×10^{-7}	1.617×10^{-7}
16	2.532×10^{-5}	3.715×10^{-5}	4.039×10^{-5}
64	6.919×10^{-4}	8.357×10^{-4}	9.081×10^{-4}

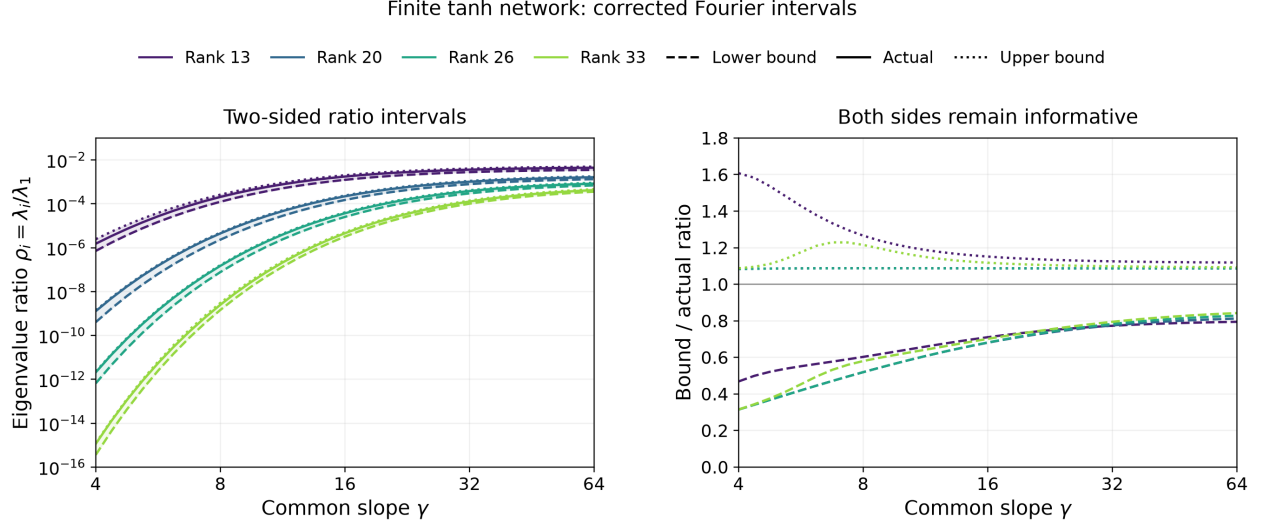


Figure 2: Actual finite-kernel ratios and the two endpoints from (10), at fixed centers and samples. Each color denotes an eigenvalue rank; solid lines are actual ratios, dashed lines the lower endpoints, and dotted lines the upper endpoints. The companion panel displays bound divided by actual ratio. These are single-gamma calculations, not comparisons anchored to an earlier spectrum. Numerical evaluations and their checks are described in Appendix F.

The intervals at gamma 4 and 8 are disjoint by more than four orders of magnitude. The Fourier integral supplies an increasing contribution, and the finite corrections and normalization retain this large improvement. This is a checked comparison for the stated geometry. Positivity of (7) alone would not prove that every finite ratio increases: the corrections and the largest eigenvalue also depend on gamma. Appendix E gives an explicit counterexample to that stronger claim.

4 The ratio interval bounds both the error and the required number of steps

For $0 \leq \rho \leq 1$, the factor $(1 - \rho/2)^{2n}$ decreases as ρ increases. Substituting the endpoints of (10) into (3) therefore gives

$$\begin{aligned} \underline{E}_\gamma(n)^2 &:= \sum_{i=1}^m p_i(\gamma) (1 - \bar{\rho}_i(\gamma)/2)^{2n}, \\ \bar{E}_\gamma(n)^2 &:= \sum_{i=1}^m p_i(\gamma) (1 - \underline{\rho}_i(\gamma)/2)^{2n}, \\ \boxed{\underline{E}_\gamma(n)^2 \leq E_\gamma(n)^2 \leq \bar{E}_\gamma(n)^2.} \end{aligned} \tag{11}$$

Both sums use the same actual finite-network weights $p_i(\gamma)$. We do not replace them by eigenvector weights from H_γ or S_γ . The ratio theorem is target-independent; applying it to a particular target still requires its projections, or justified bounds on their energy.

The lower curve says how much error must remain: if $\underline{E}_\gamma(n) > \epsilon$, no iterate up to step n can have reached tolerance. The upper curve says how much error can remain: if $\bar{E}_\gamma(n) \leq \epsilon$, the true curve has reached tolerance by that step. Define the first integer crossings

$$\underline{n}_\epsilon = \min\{n \geq 0 : \underline{E}_\gamma(n) \leq \epsilon\}, \quad \bar{n}_\epsilon = \min\{n \geq 0 : \bar{E}_\gamma(n) \leq \epsilon\},$$

with the minimum of an empty set equal to $+\infty$. Then

$$\boxed{\underline{n}_\epsilon(\gamma) \leq n_\epsilon(\gamma) \leq \bar{n}_\epsilon(\gamma)}. \quad (12)$$

The two eigenvalue bounds have different jobs: upper ratios give a necessary time, while lower ratios give a sufficient time. If a lower ratio endpoint is zero, its entire target weight remains in the upper error curve. An infinite sufficient-time bound means that this bound cannot certify convergence; it does not by itself prove that gradient descent never converges.

For the example, use

$$f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x), \quad x \in [-1, 1],$$

with $N = 128$ interior intervals, $h = 1/64$, 153 tanh centers including the halo, and 263 samples. Figure 3 applies (11) to the same geometry at five gammas. Unresolved numerical components are omitted only from the lower curve and retained in full in the upper curve; the computed finite-feature-span residual is also retained in the upper curve. Additional numerical allowances make this evaluation more conservative than the selected-rank intervals in Figure 2.

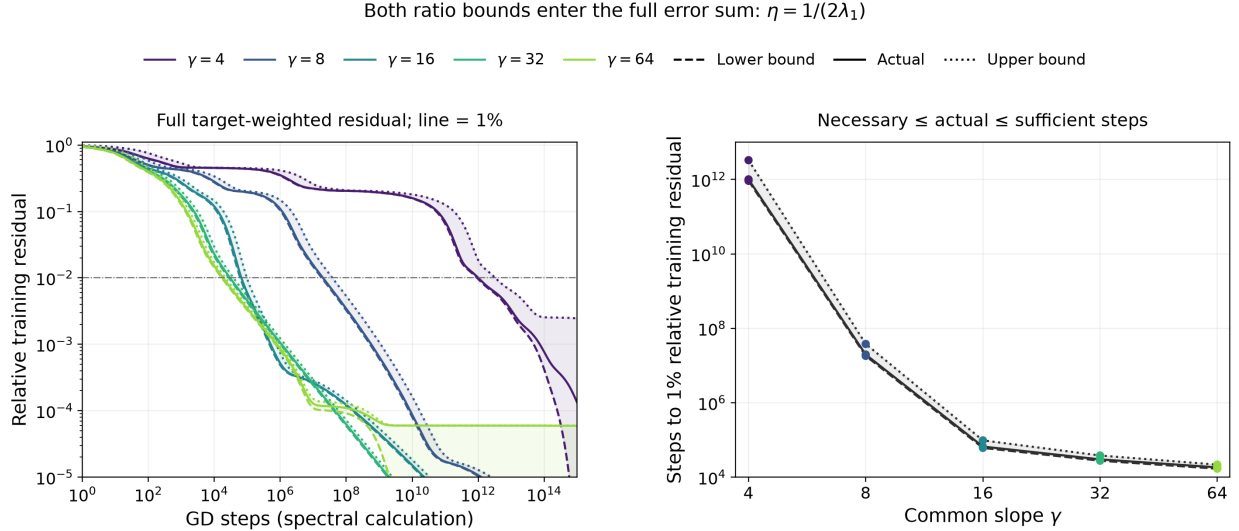


Figure 3: The finite-kernel spectral error, lower error curve using upper ratio endpoints, and upper error curve using lower endpoints. The 1% threshold determines the step-count interval. Every curve is calculated by eigendecomposition and scalar powers; no long gradient descent run was executed. The bound curves use the actual target projections and the interval theorem, rather than a separate integral-kernel forecast. Appendix F documents numerical allowances and unresolved target energy.

γ	Necessary steps	Finite-kernel spectral steps	Sufficient steps
4	9.20×10^{11}	1.008×10^{12}	3.31×10^{12}
8	18,111,765	19,697,563	37,941,446
16	60,905	66,209	97,172
32	27,547	29,938	38,036
64	16,814	18,272	21,611

The gamma-4 bound endpoints are rounded outward. These are checked floating-point evaluations of (11)–(12) with the numerical allowances in Appendix F, not certified bounds for the exact tanh kernel or guarantees for a floating-point gradient descent implementation. The necessary bound at gamma 4 is about 91% of the finite-kernel spectral count; the sufficient bound is about 3.3 times that count.

At gamma 4, a separately computed readout achieves relative error 5.85×10^{-4} , below the requested 1%. Thus the long delay is not caused by that tolerance being unrepresentable. The high accuracy exists at the same geometry; ordinary readout gradient descent needs an impractical number of steps to obtain even the much weaker 1% fit. At the larger tested gammas the relevant ratio intervals increase and the necessary and sufficient step counts decrease substantially. Appendix E extends the necessary-delay argument to stable scalar gradient descent schedules.

A The center integral and its restriction

For $d = x_a - x_b \neq 0$, the identity

$$1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$$

with $u = \gamma(x_a - c)$, $v = \gamma(x_b - c)$, gives

$$\int_{\mathbb{R}} (1 - g_{ab}(c)) dc = \coth(\gamma d) \int_{\mathbb{R}} [\tanh(\gamma(x_a - c)) - \tanh(\gamma(x_b - c))] dc = 2d \coth(\gamma d).$$

The difference of tanhs is integrable, and its log-cosh antiderivative gives $2d$. At zero separation, the integral is that of sech^2 , giving $2/\gamma$. Splitting the whole-line deficit into I_c and its complement proves (5). The cell length Wh , including half a cell beyond each outer center, is required for its constant term.

For any u with $\mathbf{1}^\top u = 0$, the constant part of (5) vanishes. In the finite double sum,

$$\int_{\mathbb{R}} \left| \sum_a u_a \tanh(\gamma(x_a - c)) \right|^2 dc = -2 \sum_{a,b} u_a u_b d_{ab} \coth(\gamma d_{ab}).$$

One can obtain this equality by integrating the products minus one; all these deficits are integrable, and the constant coefficients sum to zero. Substitution $u = Qv$ yields $H_\gamma = Q^\top K_\gamma^{(0)} Q$, proving the explicit formula used to construct the integral matrix.

The full matrix $K_\gamma^{(0)}$ need not be positive semidefinite; H_γ is, by its integral definition. The entrywise nonnegative halo-deficit matrix in (5) need not be positive semidefinite either. The positive omitted-center matrix used in (8) is a different object: it is a sum of projected feature outer products on the lattice. The proof below uses that outer-product sign, not an entrywise sign assumption.

B Fourier identity and gamma dependence

The cumulative integral of ρ_γ^{av} is $(1 + \tanh(\gamma x))/2$, proving the step-convolution identity. Substituting $s = e^{2\gamma x}$ in its transform, with $\nu = \omega/(2\gamma)$, gives

$$\widehat{\rho_\gamma^{\text{av}}}(\omega) = \int_0^\infty \frac{s^{-i\nu}}{(1+s)^2} ds = \Gamma(1-i\nu)\Gamma(1+i\nu) = \frac{\pi\nu}{\sinh(\pi\nu)}.$$

The middle equality is the beta integral and the last follows from the gamma reflection identity. The value at zero follows by continuity. For $z > 0$, $d \log(z/\sinh z)/dz = 1/z - \coth z < 0$; since z decreases as gamma increases, M_γ increases.

For $u = Qv$, the step combination $g_{\text{step}}(c) = \sum_a u_a \text{sgn}(c - x_a)$ is compactly supported because $\sum_a u_a = 0$. Its derivative is $2 \sum_a u_a \delta_{x_a}$. Thus, away from zero,

$$\widehat{g}_{\text{step}}(\omega) = \frac{2}{i\omega} \sum_a u_a e^{-i\omega x_a}, \quad \widehat{g}_\gamma = M_\gamma \widehat{g}_{\text{step}}.$$

The Fourier numerator vanishes at zero, and Parseval gives (7). No transform of a lone nondecaying tanh is needed. The two-input kernel identity in the main text can separately be understood as a tempered-distribution identity; it follows by applying the averaging density in each input.

For $\gamma_b \geq \gamma_a$, subtracting (7) at the two gammas gives a nonnegative integral for every v . Hence $H_{\gamma_b} - H_{\gamma_a} \geq 0$. Courant–Fischer then proves the ordering of their eigenvalues without differentiating or tracking eigenvectors. This order applies to the integral matrices. It does not discard the finite corrections or the normalization in (10).

C Analytic corrections for the lattice and the finite halo

Write the center lattice as $c_0 + h\mathbb{Z}$. For the zero-mean restriction, the infinite lattice sum converges. The finite sum equals that infinite sum minus the absent centers. The bias vanishes under Q .

C.1 Explicit lattice terms

Define the integrable product deficit

$$F_{ab}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c)) - 1.$$

For $\nu \neq 0$, its Fourier transform is

$$\widehat{F}_{ab}(\nu) = -\frac{4M_\gamma(\nu)}{\nu} \sin(\nu d_{ab}/2) \coth(\gamma d_{ab}) e^{-i\nu(x_a+x_b)/2}. \quad (\text{C1})$$

The diagonal limit is $-2M_\gamma(\nu)e^{-i\nu x_a}/\gamma$. The identity follows by expressing the product minus one as a coth factor times the difference of two tanhs, then transforming the integrable difference. Pairing positive and negative lattice frequencies $\nu_k = 2\pi k/h$ gives real symmetric corrections

$$G_{\gamma,k} = \frac{2}{hm} Q^\top \operatorname{Re} \left[e^{i\nu_k c_0} \widehat{F}(\nu_k) \right] Q, \quad G_\gamma = \sum_{k=1}^p G_{\gamma,k}. \quad (\text{C2})$$

The truncation count $p \geq 0$ is a computational choice, not target energy. Individual correction matrices need not be positive semidefinite. Their gamma dependence is explicit in (C1), including the multiplier at each lattice frequency.

C.2 Bound on unretained lattice terms

Choose any $0 < \vartheta < \pi/2$ and let $d = \vartheta/\gamma$. For a real zero-mean vector u , write $g(c) = \sum_a u_a \tanh(\gamma(c - x_a))$, and $\bar{x} = m^{-1} \sum_a x_a$. On the two boundaries of the analytic strip,

$$\int_{\mathbb{R}} |g(t \pm id)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \vartheta \|x - \bar{x}\mathbf{1}\|^2 \|u\|^2. \quad (\text{C3})$$

To see the constant, use $\sum u_a = 0$ to subtract $\tanh(\gamma(c - \bar{x}))$ from every term, express each difference as an integral of the derivative, and apply Minkowski and Cauchy–Schwarz. The derivative has squared L_2 norm at most $(4\gamma/3) \sec^4 \vartheta$, because $|\operatorname{sech}^2(t + i\vartheta)| \leq \sec^2 \vartheta \operatorname{sech}^2 t$ and $\int \operatorname{sech}^4 t dt = 4/3$.

Contour shifting applies to the analytic function $g(z)^2$, not to $|g(z)|^2$. Equation (C3) bounds the shifted absolute integral, so its Fourier transform decays as $e^{-d|\nu|}$. Poisson summation and a geometric series over the unretained indices $k > p$ give the operator-norm allowance

$$\delta_\gamma = \frac{8\gamma \|x - \bar{x}\mathbf{1}\|^2 \sec^4 \vartheta}{3hm [e^{2\pi\vartheta/(\gamma h)} - 1]} e^{-2\pi\vartheta p/(\gamma h)}. \quad (\text{C4})$$

This is uniform over unit vectors in the restricted space. In the calculation, $\vartheta = \arctan(\pi/(2\gamma h))$, and enough explicit terms are retained to make (C4) at most 10^{-18} . The allowance concerns uncomputed terms, not floating-point evaluation error.

C.3 Exterior centers and the matrix enclosure

Let $\mathcal{F}$ be a finite collection of absent lattice centers, and retain their contribution

$$T_\gamma = \frac{1}{m} \sum_{c \in \mathcal{F}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q.$$

Include all absent centers inside the sample range. Suppose the unretained centers begin at $c_R > x_{\max}$ and $c_L < x_{\min}$. The inequality $1 - \tanh t \leq 2e^{-2t}$ for $t \geq 0$, cancellation of the constant vectors under Q , and summation over the two exterior lattice tails give

$$0 \leq T_{\text{tail},\gamma} \preceq \tau_\gamma I, \quad \tau_\gamma = \frac{4[e^{-4\gamma(c_R - x_{\max})} + e^{-4\gamma(x_{\min} - c_L)}]}{1 - e^{-4\gamma h}}. \quad (\text{C5})$$

For example, each right-tail projected outer product has norm at most $4e^{-4\gamma(c - x_{\max})}$ after the factor $1/m$; the geometric series gives its contribution to (C5). The implementation retains $\lceil 18/(\gamma h) \rceil$ absent centers on each side of the finite network.

Poisson summation gives the infinite lattice matrix as $H_\gamma + G_\gamma + R_\gamma$, where $\|R_\gamma\|_2 \leq \delta_\gamma$. Subtracting all absent centers therefore gives exactly

$$Q^\top K_\gamma Q = S_\gamma + R_\gamma - T_{\text{tail},\gamma}.$$

This proves (9). All quantities are evaluated at the same gamma. The retained corrections may be substantial; it is their unretained remainder that must be bounded. A small global error relative to the largest eigenvalue is not enough when studying a much smaller eigenvalue.

D Eigenvalue endpoints and normalization

Let $\kappa_1 \geq \dots \geq \kappa_{m-1}$ be the eigenvalues of $Q^\top K_\gamma Q$. Weyl's order inequalities applied to (9) give

$$\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma.$$

Compression interlacing gives $\lambda_i \geq \kappa_i \geq \lambda_{i+1}$. Combining the lower inequality at index i and the upper inequality at index $i - 1$ proves (10). For the last full eigenvalue, only the upper interlacing inequality is available; nonnegativity supplies its lower bound. Since $\text{rank } B_\gamma \leq W + 1$, all indices $i > W + 1$ are exact zeros, if such indices exist. Numerically small positive singular values are not thereby exact zeros.

For the largest eigenvalue, set $e = 1/\sqrt{m}$. Its Rayleigh quotient supplies

$$\ell_\gamma = e^\top K_\gamma e = \|B_\gamma^\top e\|^2 = 1 + \sum_j \left[\frac{1}{m} \sum_a \tanh(\gamma(x_a - c_j)) \right]^2. \quad (\text{D1})$$

In the orthonormal basis $[e, Q]$, the kernel has blocks $\begin{pmatrix} \ell_\gamma & b_{\text{vec}}^\top \\ b_{\text{vec}} & Q^\top K_\gamma Q \end{pmatrix}$, where $b_{\text{vec}} = Q^\top B_\gamma (B_\gamma^\top e)$. Put $b = \|b_{\text{vec}}\|$ and $c = \beta_1 + \delta_\gamma$. Bounding the cross term by b and the restricted block by cI yields

$$L_\gamma = \frac{\ell_\gamma + c + \sqrt{(\ell_\gamma - c)^2 + 4b^2}}{2} \geq \lambda_1(K_\gamma). \quad (\text{D2})$$

This is the largest eigenvalue of a two-by-two scalar matrix. Equations (D1)–(D2) use finite-feature matrix-vector products and the corrected integral spectrum. They do not use the actual finite small eigenvalues. The actual feature SVD is used separately to compare the bounds and to

obtain the target weights for Section 4. Coarser universal bounds follow from the bias and trace: $1 \leq \lambda_1 \leq W + 1$.

E What the training bounds imply, and what they do not

E.1 Error bounds and threshold crossings

For every index, $0 \leq \underline{\rho}_i \leq \rho_i \leq \bar{\rho}_i \leq 1$. Multiplying the corresponding scalar inequalities by nonnegative p_i and summing proves (11). All three curves are nonincreasing in n , so the first crossing of the lower curve cannot be later than the actual crossing, and the first crossing of the upper curve cannot be earlier. This proves (12), including empty-set cases.

Omitting nonnegative terms preserves a lower error bound. It does not preserve an upper error bound. If some component is unresolved, assigning it lower rate zero retains its full weight in the upper curve. Exact zero-eigenvalue components have factor one in the actual curve; they must not be confused with unresolved positive eigenvalues. Repeated eigenvalues cause no ambiguity in the actual curve because the sum of weights in their eigenspace is basis-independent.

A compact necessary-time test does not require the full error curve. Let $P_i^+ = \sum_{j \geq i: \lambda_j > 0} p_j$, the target's squared-norm fraction in positive-eigenvalue directions at ranks $j \geq i$. Ordering gives $\rho_j \leq \rho_i \leq \bar{\rho}_i$, and hence

$$E_\gamma(n)^2 \geq P_i^+(1 - \bar{\rho}_i/2)^{2n}.$$

If $P_i^+ > \epsilon^2$ and $\bar{\rho}_i > 0$, attaining tolerance requires

$$n_\epsilon \geq \left\lceil \frac{\log(\sqrt{P_i^+}/\epsilon)}{-\log(1 - \bar{\rho}_i/2)} \right\rceil. \quad (\text{E1})$$

More generally, a known lower bound on P_i^+ , or the energy of any subset of these positive directions, gives a valid necessary-time bound. Positivity matters for the interpretation: $u_j = B_\gamma(B_\gamma^\top u_j / \lambda_j)$ when $\lambda_j > 0$, so that component is representable. The theorem does not supply the target weights from gamma alone.

E.2 Scalar learning-rate schedules

Keep the same frozen kernel, but allow $0 \leq \eta_t \leq 2/\lambda_1$ at every step. If a selected positive band has ratios at most $b < 1/2$, then $1 - \eta_t \lambda_j \geq 1 - 2b > 0$. A target energy fraction P in that band therefore gives

$$E(n)^2 \geq P(1 - 2b)^{2n}, \quad n_\epsilon \geq \left\lceil \frac{\log(\sqrt{P}/\epsilon)}{-\log(1 - 2b)} \right\rceil \quad (P > \epsilon^2).$$

For small b , allowing this schedule weakens the displayed necessary-time lower bound by approximately a factor of four compared with $\eta = 0.5/\lambda_1$. This is a comparison of the lower bounds, not a general theorem about the ratio of actual crossing times. It does not cover steps outside the stated per-update stability range, momentum, or matrix preconditioning.

These results concern frozen geometry and Euclidean readout gradient descent, not Adam, preconditioned readout coordinates, or the joint dynamics of gamma.

E.3 Why the theorem does not claim universal ratio monotonicity

Take samples $(-1, 1)$, centers $(-3/2, 0, 3/2)$, and the bias. Both sets are uniform and symmetric. Put $a = \tanh(5\gamma/2)$, $b = \tanh(\gamma/2)$. The even and odd eigenvalues of the normalized kernel are

$$\lambda_+ = 1 + \frac{1}{2}(a + b)^2, \quad \lambda_- = \tanh^2 \gamma + \frac{1}{2}(a - b)^2, \quad \lambda_+ > \lambda_-.$$

As $\gamma \rightarrow \infty$,

$$\frac{\lambda_-}{\lambda_+} = \frac{1}{3} + \frac{4}{9}e^{-\gamma} + O(e^{-2\gamma}).$$

The ratio eventually decreases. Thus a universal assertion that increasing gamma always improves every finite ratio would be false. The note instead provides absolute bounds at each gamma and demonstrates disjoint intervals and faster spectral convergence in the specified regimes. Its small-gamma budget obstruction does not require that stronger assertion.

F Numerical calculations and reproducibility

F.1 Geometry, target, and scope

The calculations use $N = 128$ equal intervals on $[-1, 1]$, $h = 2/N$, $N + 1 = 129$ interior centers, and $\lceil \sqrt{N} \rceil = 12$ halo centers per side. There are $W = 153$ tanh neurons and one bias feature. The 263 samples include both endpoints and are equally spaced. The center lattice runs from -1.1875 to 1.1875 . Every gamma uses the same samples, centers, target, and raw readout parameterization. The learning rate is normalized separately by each actual λ_1 .

The selected-rank plot reuses 33 logarithmically spaced gammas from 4 to 64, at ranks 13, 20, 26, and 33, including both symmetry classes. The full error and time intervals are evaluated at gamma 4, 8, 16, 32, and 64. The mixed-sine target is the one stated in Section 4. The result concerns training error, not a claim about error between samples.

F.2 Computation and numerical allowances

The integral matrix is evaluated through a positive quadrature feature factor, using piecewise Gauss–Legendre quadrature. Its SVD supplies a basis in which the integral spectrum is diagonal; the explicit lattice corrections and omitted-center outer products are added in that basis. This avoids subtracting nearly equal large entries before resolving small eigenvalues. The finite-feature singular values are squared to obtain the actual eigenvalues, rather than diagonalizing a rounded Gram matrix.

The exact theorem uses the whole-line H_γ . Truncating its integration domain at $\max_a |x_a| + P/\gamma$ omits a positive matrix with norm at most $2e^{-4P}/(h\gamma)$. The implementation adds that allowance to upper eigenvalue endpoints and the upper normalization. The exterior lattice remainder is separately controlled by (C5).

The analytic allowances do not include quadrature or floating-point error. Figure 2 shows the original selected-rank evaluations, checked by increasing quadrature order from 10 to 16 and integration padding from 20 to 24, and by independent feature-SVD drivers. The maximum checked relative change in the selected corrected eigenvalues was 5.8×10^{-8} ; actual finite-feature eigenvalues agreed to 7.2×10^{-12} relative at those ranks. Those are numerical checks, not certified bounds on every rounding error.

For the full error curves, add a numerical allowance $\nu_\gamma = 64\epsilon_{64}\|S_\gamma\|_2$, where ϵ_{64} is double-precision machine epsilon. Subtract it from lower eigenvalue numerators; add it to upper numerators and the restricted-block upper bound in (D2). This empirical allowance is checked against independent

refinement, not proved to enclose every rounding error. It prevents numerical positive eigenvalues near the precision limit from supporting unjustified positive lower rates.

Actual feature-SVD ratios above 10^{-18} are retained as numerically resolved. Every other mode receives zero lower rate and keeps its full energy in the upper error curve. The lower curve drops both the unresolved-mode energy and the computed projection remainder outside the returned finite-feature singular vectors. The upper curve retains both. The projection remainder is recorded separately from unresolved positive-mode energy; neither numerical split is used to certify the true asymptotic nullspace floor.

At all five gammas, the guarded endpoints bracket every retained numerical ratio, and the full error curves bracket the finite-kernel spectral curve. Repeating the construction with quadrature order 16 and padding 24 changes the gamma-4 bound counts by less than 8×10^{-9} relatively; the other bound counts agree as integers. After changing basis, the corrected-matrix refinement discrepancy is below the chosen numerical allowance. Independent feature-SVD drivers also preserve the resolved-mode ordering and count intervals. The upper curves all cross 1%; their nondecaying energy is explicitly retained, not discarded to force a finite crossing. The crossing search is capped at 10^{18} ; no reported count hits that cap.

For very small rates, powers are evaluated as $\exp(2n \log 1p(-\rho/2))$. Integer crossing times use bracketing and binary search. These calculations evaluate the frozen-kernel recurrence in exact-arithmetic form from computed spectral data; trillion-step counts do not imply that trillion-step floating-point training was executed or verified.

F.3 Proven statements and numerical evidence

Statement	Status
gradient descent recurrence and target-weighted error formula	Exact for the specified frozen-feature updates and zero initial readout.
Integral decomposition, Fourier identity, and ordered growth of H_γ	Proved under the stated definitions; no Fourier-eigenvector assumption.
Corrected finite ratio interval	Proved with analytic lattice and exterior-center allowances.
Error sandwich, first-crossing interval, and positive-tail delay	Proved using the same actual target weights or justified energy bounds.
Evaluated intervals, attainable readouts, and step-count numbers	Checked floating-point results; no directed-rounding certificate.
Every finite ratio or target crossing improves with gamma	Not claimed; universal ratio monotonicity is false.

F.4 Provenance and relation to the paper draft

This edition preserves the four-part argument and the complete proof appendices from the revision-4 gamma-ratio note (23 September 2026), together with its existing figures. Its two independent revision-4 reviews specify the treatment of target weights, unresolved modes, interlacing indices, and numerical allowances. The original sources are:

- docs/frozen_geometry_capacity_access/gamma_ratio_note/gamma_ratio_note_v4.md; companion files revision4_math_review.md and revision4_argument_review.md in that directory.
- The experiment sources direct_ratio_interval.py and note_interval_figures.py under experiments/expD37_capacity_access_figures/gamma_solve/spectrum_mechanism/.

- Saved reports and figure manifests under the matching `results/checkpoint_D_optimizers/expD37_capacity_access_figures/gamma_solve/spectrum_mechanism/` directory, especially `direct_ratio_interval/`, `note_interval_figures/`, and `spectrum_mechanism_results.md`.

These are existing numerical results; this edition does not run new training. The example here has 153 hidden neurons, 263 samples, and a highest target frequency of 10π . It is not the current paper's separate width-512, 14π example. The figure and step-count claims in this note should retain their own experimental configuration when reused.